

TaskFlow

Team To-Do Application

Project Plan Presentation (PPP)

Agenda

1. Team Introduction
2. Problem Domain & Motivation
3. Our Solution
4. Features Overview
5. Requirements & Tests
6. Architecture
7. Burndown Metrics
8. Q&A

Meet the Team

Member	Role	Features Assigned
Denver	Team Leader	F1: Core To-Do List, F2: Sub-Tasks + Progress
Developer	Developer	F3: Kanban Columns, F4: Calendar View
Nia	Developer	F5: Canvas API, F6: Gamification

The Problem

Students struggle to manage academic workloads

- Students often **miss deadlines** due to poor task visibility
- Large assignments with **multiple sub-tasks** are hard to track
- No easy way to see **workload across all classes** at a glance
- **Lack of motivation** to complete tasks on time
- Existing apps are too **generic** or too **complex**

Why It Matters

Students who use task management tools effectively are **more likely to succeed academically**

Current Solutions Fall Short:

- Generic to-do apps lack academic-specific features
- No integration with learning management systems (Canvas)
- Missing progress visualization for multi-part assignments
- No gamification to encourage task completion

Our Solution: TaskFlow

A student-oriented task manager with:

- Sub-task progress tracking
- Kanban-style workflow management
- Calendar visualization
- Canvas LMS integration
- Gamification system

Goal: Help students stay organized, motivated, and succeed academically

Features Overview

Feature 1: Core To-Do List Management

Assigned to: Denver

ID	User Story
R1.1	As a user, I should be able to create a new task with a title and description
R1.2	As a user, I should be able to edit and delete tasks
R1.3	As a user, I should be able to mark tasks as complete

Acceptance Tests:

- Create task → verify it appears in list
- Edit/Delete task → verify changes
- Mark complete → verify status changes

Feature 2: Sub-Tasks + Progress Tracking

Assigned to: Denver

ID	User Story
R2.1	As a user, I should be able to add sub-tasks to a parent task
R2.2	As a user, I should be able to mark sub-tasks as complete independently
R2.3	As a user, I should be able to see a progress bar based on completed sub-tasks

Acceptance Tests:

- Add sub-task → verify nested display
- Complete sub-task → verify parent unchanged
- Complete 2/4 sub-tasks → verify 50% progress bar

Feature 3: Kanban Columns

Assigned to: Developer

ID	User Story
R3.1	As a user, I should be able to mark a task as "In Progress"
R3.2	As a user, I should be able to drag tasks between columns
R3.3	As a user, I should be able to see my current focus task at a glance

Columns: To-Do → In Progress → Done

Feature 4: Calendar View (Read-Only)

Assigned to: Developer

ID	User Story
R4.1	As a user, I should be able to view tasks on a calendar based on due dates
R4.2	As a user, I should be able to click a calendar task to see its details

Acceptance Tests:

- Create task with due date Feb 15 → verify on calendar
- Click task → verify details modal appears

Feature 5: Canvas API Integration

Assigned to: Nia

ID	User Story
R5.1	As a user, I should be able to connect my Canvas LMS account
R5.2	As a user, I should be able to import assignments from Canvas as tasks
R5.3	As a user, I should be able to sync Canvas assignment updates automatically

Key Value: Automatic import of assignments with due dates!

Feature 6: Points/Gamification System

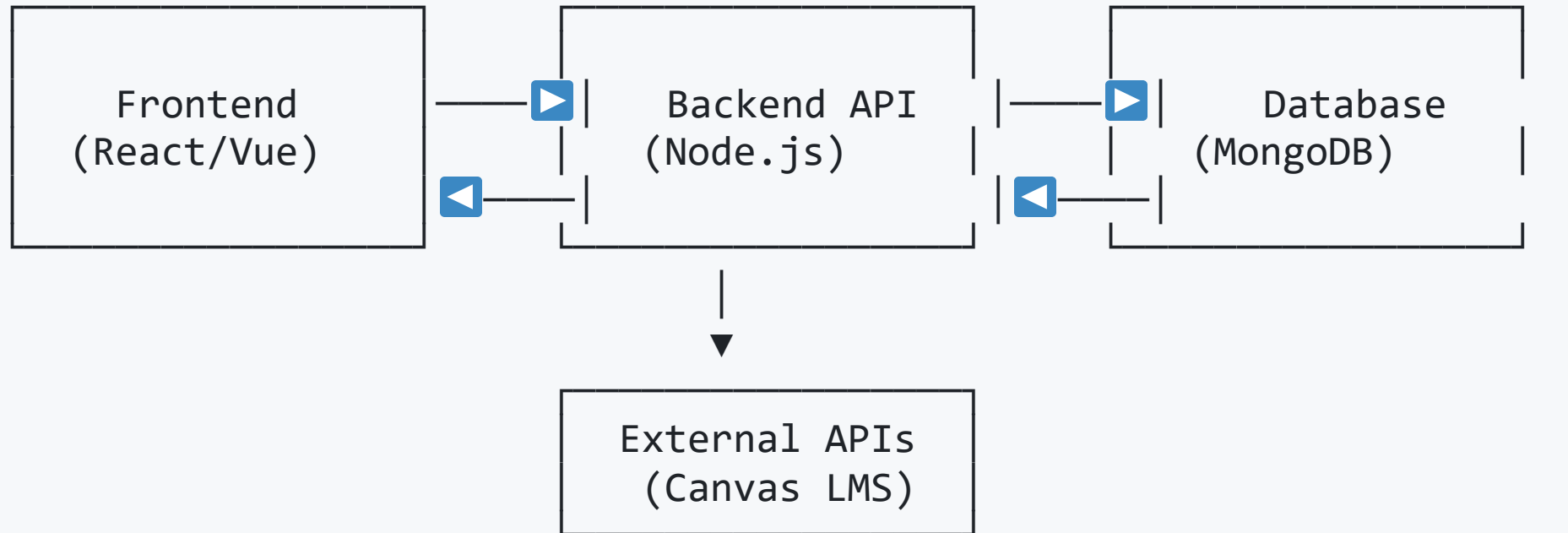
Assigned to: Nia

ID	User Story
R6.1	As a user, I should earn points when I complete tasks
R6.2	As a user, I should see my total points and level on my profile
R6.3	As a user, I should earn bonus points for completing tasks before the due date

Motivation through rewards!

Architecture

System Architecture



Data Model

Entity	Description	Key Fields
User	Application user	id, email, password_hash, points, level
Task	Individual task item	id, title, description, status, due_date, user_id
SubTask	Child task of a Task	id, title, completed, task_id

Test Strategy

Three Levels of Testing:

1. Unit Tests

- Individual function and component testing

2. Integration Tests

- API endpoint and database interaction testing

3. Acceptance Tests

- End-to-end user workflow testing

Each requirement has a corresponding acceptance test.

Burndown Metrics

Project Metrics Summary

Metric	Count
Features	6
Requirements	17
Tests	17

Feature Distribution:

- Denver: 2 features (6 requirements)
- Developer: 2 features (5 requirements)
- Nia: 2 features (6 requirements)

Requirements Breakdown by Feature

Feature	Requirements	Tests
F1: Core To-Do List	3	3
F2: Sub-Tasks + Progress	3	3
F3: Kanban Columns	3	3
F4: Calendar View	2	2
F5: Canvas API	3	3
F6: Gamification	3	3
Total	17	17

Project Links

Resource	Link
GitHub Repository	github.com/joseph-ampfer/ASE285_team_project
Canvas Team Project Page	[Link TBD]
Canvas Progress Page	[Link TBD]
Documentation	docs/
Source Code	src/
Tests	tests/

Questions?

Thank you!

GitHub: github.com/joseph-ampfer/ASE285_team_project